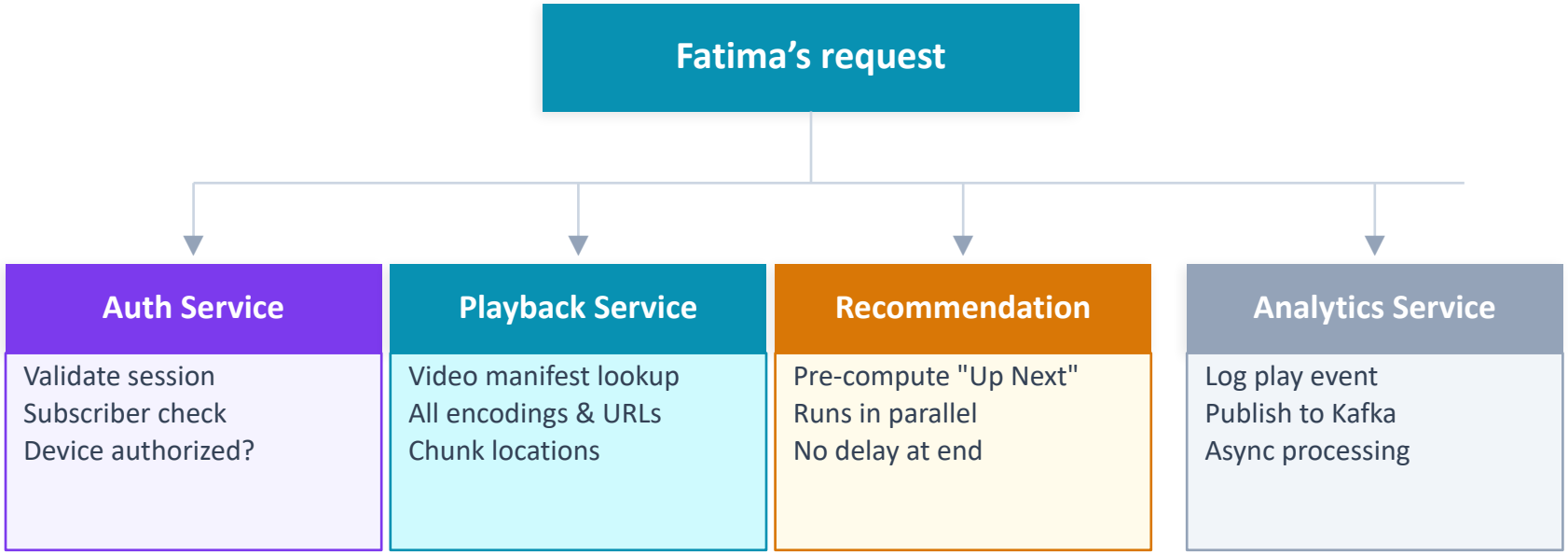
Layer 3

Microservices Fan-Out

One tap, four parallel service calls

Parallel Fan-Out



Key insight: total wait = slowest call, not the sum of all calls. All 4 fire simultaneously via gRPC.

Thundering Herd & Load Shedding

"Paws of Fury" Season 2 drops at 9pm

10 million users hit play within 30 seconds.
Each request fans out to 4 services = 40M internal calls.

Prioritized Load Shedding

PLAY REQUESTS	High Priority
RECOMMENDATIONS	Medium
ANALYTICS EVENTS	Low — can queue

Retry Storm Problem

Service overloaded → returns errors → all clients retry → doubled load on a drowning service

Mitigation

Exponential backoff
Each retry waits longer than the last

Jitter
Randomize wait times so retries don't sync

Layer 4

Data & Caching

400 million cache operations per second

Two-Tier Data Architecture

PetCache (Memcached)

200 clusters • 20,000 instances • trillions of items • sub-millisecond reads

✓ Cache Hit (fast path)

"Mittens" is popular —
manifest returned in <1ms

✗ Cache Miss (fallthrough)

Obscure title → query
falls to persistent store

▼

Apache Cassandra

Many clusters • petabytes of data • multi-region replication • eventual consistency

Consistent hashing: adding/removing a server only reshuffles 1/N of keys, not all of them

The Consistency Problem

Fatima's Watch Later list

Watch Later list appears empty!

The Consistency Spectrum

Eventual	Session-Level	Strong (ACID)
Write returns before replication. Other replicas catch up... eventually.	Fatima always sees her own writes. Others may see stale data briefly.	Write synchronously replicates everywhere before returning success.
Analytics, view counts	*Watch Later list, profiles*	*Billing, payments*

← Faster, less safe

Slower, more safe →